

AI-Powered Smart Career Guidance Assistant using Prompt Engineering

ABSTRACT

The AI-Powered Smart Career Guidance Assistant using Prompt Engineering is an intelligent web-based application designed to provide personalized career guidance for students and job seekers. Many individuals face difficulties in selecting suitable career paths, identifying required skills, and planning their learning journey. Traditional career guidance methods are often generic and lack personalization. This project addresses these challenges by integrating Artificial Intelligence and Prompt Engineering techniques to generate accurate and structured career recommendations.

The system is developed using Python, Streamlit, and AI APIs through OpenRouter. The application collects user information such as skills, interests, and career goals, and generates personalized career suggestions, learning roadmaps, industry insights, and career advice.

Additionally, the system contains a Resume Analyzer module and an AI Career Chatbot for interactive career assistance.

INTRODUCTION

Career guidance plays an important role in helping students and job seekers select suitable career paths based on their skills, interests, and goals. Artificial Intelligence and Prompt Engineering technologies are capable of generating intelligent and context-aware responses.

The AI-Powered Smart Career Guidance Assistant is designed to provide intelligent career recommendations, learning roadmaps, salary insights, and career advice using engineered prompts and AI models.

PROBLEM STATEMENT

Students and job seekers often struggle to identify suitable career paths based on their skills and interests. Traditional guidance systems lack personalization and updated industry insights.

OBJECTIVES

1. Provide personalized career recommendations.
2. Implement prompt engineering techniques.
3. Generate learning roadmaps.
4. Analyze resumes and suggest improvements.
5. Develop an AI career chatbot.

EXISTING SYSTEM

Existing career guidance systems are mostly manual, static, and non-personalized. They do not effectively utilize AI or prompt engineering techniques.

PROPOSED SYSTEM

The proposed system uses Python, Streamlit, OpenRouter APIs, and Prompt Engineering to provide personalized AI-based career guidance, resume analysis, and interactive chatbot assistance.

TECHNOLOGIES USED

- Python
- Streamlit
- OpenRouter API
- Prompt Engineering
- VS Code

FUTURE ENHANCEMENTS

1. PDF Resume Upload
2. Voice Assistant
3. AI Interview Preparation
4. Job Recommendation System

CONCLUSION

The project successfully demonstrates how Prompt Engineering and Artificial Intelligence can be used to provide intelligent and personalized career guidance for students and job seekers.